MagicCode Super-Resolution Software Getting Started

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1. Choose an integration path

Graphic SR (graphic super-resolution) uses rendered content as its input, for example games or virtual humans. Image SR (image super-resolution) uses real camera-captured content as its input, for example photos or video.

Path	Library	Header	Use when
Image (CPU or GPU)	libmagic_sr.a / libmagic_sr.lib	mc_interface.h	mc_nscaler_enable per frame with in2[2]; NULL-handle create defaults to 2x SR_SPEED_MODE_FOR_IMAGE
Graphic (GPU)	libmagic_sr.a / libmagic_sr.lib	mc_interface.h	ctrl_param_t create + temporal_frame_t per frame; in2[2]
Linux x86_64	lib/linux/libmagic_sr.a / .so	mc_interface.h	CPU image (AVX2) + Vulkan Graphic (Vulkan Temporal)/TAAU + FG
Frame generation (Vulkan)	same libmagic_sr	mc_interface.h	GEN_* alg_mode; in2[1] set for interpolate; weights for GEN_BALANCED

2. Image quick start (nscaler)

Link libmagic_sr, include mc_interface.h, bind per-frame input/output, then (a NULL handle creates a default 2x SR_SPEED_MODE_FOR_IMAGE session):

void* h = NULL; mc_nscaler_input_frame_t *in2[2] = { &in_frame, NULL }; mc_nscaler_enable(&h, in2, &out_frame); /* then mc_nscaler_disable(h) */

- Scale [1.0, 8.0]; <=0 defaults to 2.0.
- Android GLES/Vulkan: bind width/height on mc_nscaler_input_frame_t; the first enable sets the handle size (no size-hint call in v3.0.0).
- v2.2.0 removed the Enable archive and mc_enable.h. Non-default backend/scale/model: create with mc_nscaler_control(MC_NSCALER_CMD_SET_PARAM).

3. Graphic session (v3.0.0)

Link libmagic_sr, include mc_interface.h, create with mc_nscaler_control(MC_NSCALER_CMD_SET_PARAM) on a ctrl_param_t (alg_mode = SR_SPEED_MODE_FOR_GRAPHIC or SR_BALANCED_MODE_FOR_GRAPHIC, gpu_context, create-stage depth/HDR), then each frame fill mc_nscaler_input_frame_t / mc_nscaler_output_frame_t plus a temporal_frame_t (struct_size, depth, motion, jitter, frame_index, camera) and call mc_nscaler_enable with in2[2] = { &in, NULL }.

Graphic Super-Resolution

Graphic super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x SR_SPEED_MODE_FOR_IMAGE) and processes the first frame. Pass mc_nscaler_input_frame_t *in2[2] = { &in, NULL } (set in2[1] only for GEN interpolate).

Modes

Value	Enumerator	Role
0	SR_SPEED_MODE_FOR_IMAGE	Image, throughput first
1	SR_BALANCED_MODE_FOR_IMAGE	Image, quality/cost trade-off
2	SR_SPEED_MODE_FOR_GRAPHIC	Graphic, throughput first
3	SR_BALANCED_MODE_FOR_GRAPHIC	Graphic, canonical FSR-family path
4	GEN_SPEED_MODE_FOR_GRAPHIC	FG-Fast (analytic, no weights). Vulkan + MAGIC_SR_HAS_FG. Combinable with SR bits; bare 4 equals SR_SPEED_MODE_FOR_IMAGE|GEN_SPEED.
8	GEN_BALANCED_MODE_FOR_GRAPHIC	FG-CoreV3. Same availability as Fast. Needs magic_fg_core.bin. Do not set both GEN bits.

- Graphic scaler_factor is (1, 8]; 1.0 itself is rejected. Image remains [1.0, 8.0].
- SR_SPEED_MODE_FOR_GRAPHIC on Metal, Vulkan, and OpenGLES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family graphic path (same family as Balanced, not a separate customer API).
- SR_BALANCED_MODE_FOR_GRAPHIC uses the canonical FSR-family path on every GPU graphic backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLES do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run graphic modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at create (mc_nscaler_control SET_PARAM, or the first mc_nscaler_enable): Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan graphic requires mc_nscaler_input_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on ctrl_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both graphic modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

3.1 Frame generation (optional, Vulkan)

Add GEN_SPEED_MODE_FOR_GRAPHIC or GEN_BALANCED_MODE_FOR_GRAPHIC to alg_mode. Pass in2[1]=NULL for SR, or a second filled frame to interpolate t=0.5. Vulkan only; Windows/Linux product libraries define MAGIC_SR_HAS_FG. Android Vulkan currently does not. GEN_BALANCED needs model/magic_fg_core.bin (see Integration Guide).

- Vulkan Speed (Lite-TAAU) now lets an explicit strong reactive mask suppress the still-camera accumulation branch. Previously that branch ignored the caller reactive mask when the camera was still or per-pixel MV was below about 0.2 px, so pixels with ~0 MV but changing content (TV / video-wall / LED-screen textures) could keep ghosting for about 64–96 frames after a content cut. Strong reactive (≥ about 0.85, fully effective at ≥ 0.95) now weakens that

accumulation. Omitting reactive or passing a weak value (< 0.85) is unchanged. Recommendation: mark screen-like objects whose MV is 0 but whose content changes (for example with CustomDepth stencil) as reactive at strength 1.0. Limit: Vulkan only in this release. The Metal / GLES / D3D11 Speed still-accumulation structure is the same idea but is not modified or verified here.

4. Models

GPU model bins are magic_<backend>_speed_gpu_params.bin and magic_<backend>_balanced_gpu_params.bin. Use tools/setup_models.sh (copies Speed + Balanced GPU bins). Set ctrl_param_t.model_path at create in production apps. Linux CPU image uses the CPU model bin (magic_sr_cpu_params.bin). GEN_BALANCED also needs ctrl_param_t.model_path set to the FG weight file (typically model/magic_fg_core.bin; empty uses MAGIC_FG_WEIGHT_PATH then default search). Recreate the handle to change weights.

5. Linux platform library

- lib/linux/libmagic_sr.a and libmagic_sr.so ship x86_64 CPU image (AVX2/FMA, POSIX threads) plus Vulkan Graphic (Vulkan Temporal)/TAAU and FG. Rebuild with tools/build_linux_lib.sh; link with -lm -lpthread.
- Linux Vulkan Graphic (Vulkan Temporal)/TAAU and FG objects are in the product archive (MAGIC_ENABLE_VULKAN_TEMPORAL, MAGIC_ENABLE_VULKAN_TAAU, MAGIC_SR_HAS_FG).
- CPU graphic is not available on any platform, including Linux.

6. Next documents

- User Guide: doc/User_Guide.md and doc/用户使用说明书.md
- API Reference, Integration Guide, FAQ, Migration Guide, Known Issues in this folder